

Scientific Computation

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Lecture 7 - Linear Algebra Refresher

Lecture Outline

- Vector spaces
- Matrices

- Linear Systems
- Eigenvalues
- Orthogonality

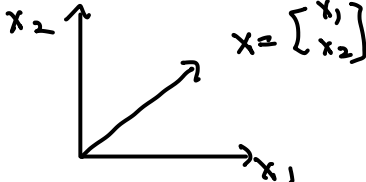
Vector Spaces

For a field $\mathbb{K}$ (in this course, $\mathbb{K} = \mathbb{R}$), a vector space V with binary addition operator $+$ satisfies the following axioms for every vector $u, v, w \in V$ and scalar $a, b \in \mathbb{K}$:

1. $u + (v + w) = (u + v) + w$ (associativity of vector addition)
2. $u + v = v + u$ (commutativity of vector addition)
3. $\exists 0 \in V$ such that $v + 0 = v$ (existence of zero vector)
4. $\exists -v \in V$ such that $v + (-v) = 0$ (additive inverse)
5. $a(bv) = (ab)v$ (compatibility of scalar and field multiplication)
6. $1v = v$ where 1 is the multiplicative identity in $\mathbb{K}$
7. $a(u + v) = au + av$ (distributivity with respect to vector addition)
8. $(a + b)v = av + bv$ (distributivity with respect to field addition)

Euclidean Space

In this course, we deal almost exclusively with n -dimensional Euclidean space $\mathbb{R}^n$, which is a vector space over the field $\mathbb{R}$ (show that this obeys the axioms of a vector space):

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{where } x_i \in \mathbb{R} \text{ for } i=1 \dots n$$


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Real Matrices

Consider a real matrix $A \in \mathbb{R}^{m \times n}$.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & & \\ \vdots & & \ddots & \\ a_{m1} & & & a_{mn} \end{bmatrix}$$

$\downarrow$ m rows $\rightarrow$ n cols

If $m=n$, we have
a square matrix

$$A^T = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & \dots & & \\ \vdots & & \ddots & \\ a_{1n} & & & a_{mn} \end{bmatrix}$$

where each $a_{ij} \in \mathbb{R}$

Transpose

Sometimes written $A_{ij} = (A)_{ij}$

6

Matrix-Vector Multiplication

A is not just an array of numbers!

We multiply a real matrix $A \in \mathbb{R}^{m \times n}$ with a compatible vector $x \in \mathbb{R}^n$:

$$(Ax)_i = \sum_{j=1}^n A_{ij} x_j, \quad Ax \in \mathbb{R}^m$$

$\left[\begin{smallmatrix} \text{---} & \text{---} & \text{---} \end{smallmatrix} \right] \cdot \left[\begin{smallmatrix} \text{---} \\ \text{---} \\ \text{---} \end{smallmatrix} \right]$

$\left[\begin{smallmatrix} \text{---} & \text{---} & \text{---} \end{smallmatrix} \right] \cdot \left[\begin{smallmatrix} \text{---} \\ \text{---} \\ \text{---} \end{smallmatrix} \right]$

$$\text{Let } A = \begin{bmatrix} -a_1^T \\ \vdots \\ -a_m^T \end{bmatrix}, \text{ then } Ax = \begin{bmatrix} \vdots \\ a_m^T x \end{bmatrix}$$

↳ row vec of A

$$x^T y = x \cdot y = \sum_{i=1}^n x_i y_i$$

vectors are
 $\mathbb{R}^{n-1}$ matrices

Interpret A as a linear
mapping: $A: \mathbb{R}^n \rightarrow \mathbb{R}^m$

7

Matrix-Matrix Multiplication

We can also multiply a real matrix $A \in \mathbb{R}^{m \times n}$ with a compatible matrix $B \in \mathbb{R}^{n \times p}$:

Do matrix-vector mult. on each column
of B :

$$AB = \begin{bmatrix} -a_1^T \\ \vdots \\ -a_m^T \end{bmatrix} \begin{bmatrix} | & & | \\ b_1 & \dots & b_p \\ | & & | \end{bmatrix} = \begin{bmatrix} a_1^T b_1 & \dots & a_1^T b_p \\ \vdots & \ddots & \vdots \\ a_m^T b_1 & \dots & a_m^T b_p \end{bmatrix}$$

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_i^T b_j \\ = a_i \cdot b_j$$

$$= [Ab_1 \quad Ab_2 \quad \dots \quad Ab_p] \\ \in \mathbb{R}^{m \times p}$$

8

Lecture Outline

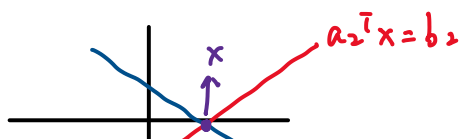
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- **Linear Systems**
- Eigenvalues
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Linear Systems of Equations

This section of the course is all about solving $Ax = b$ given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ for some unknown vector $x \in \mathbb{R}^n$:

$$\begin{bmatrix} -a_1^T \\ \vdots \\ -a_m^T \end{bmatrix} x = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}, \quad \begin{matrix} m \\ \text{Eqs.} \end{matrix} \quad \begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

E.g. $m=n=2$





$$a_1^T x = b_1$$

10

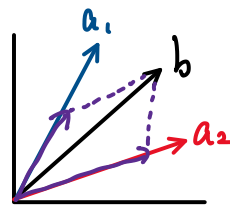
Alternate Interpretation

This section of the course is all about solving $Ax = b$ given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ for some unknown vector $x \in \mathbb{R}^n$:

$$\begin{bmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + x_2 a_2 + \dots + x_n a_n = b$$

not the same
as a_1 in the
last slides, these
are column vecs
here!

E.g. $m=n=2$



x_1 and x_2 to reach
 b with $\vec{a}_1$ and $\vec{a}_2$

11

Subspaces

for any vec space, not just $\mathbb{R}^n$

A linear subspace $V \subseteq \mathbb{R}^n$ is a subset that is a vector space itself:

Using the same addition and scalar field
multiplication operations, we can show that V

is also a vector space.

Equivalently: V is a subspace of $\mathbb{R}^n$ if $\alpha v_1 + \beta v_2 \in V$ for any $v_1, v_2 \in V$ and $\alpha, \beta \in \mathbb{R}$

Ex. $V = \left\{ \begin{bmatrix} \alpha \\ \alpha \end{bmatrix}, \alpha \in \mathbb{R} \right\} \in \mathbb{R}^2$ is a linear subspace

show that $W = \left\{ \begin{bmatrix} \alpha+1 \\ \alpha \end{bmatrix}, \alpha \in \mathbb{R} \right\} \in \mathbb{R}^2$ not linear subspace

12

Nullspace

An important subspace is the **nullspace** or **kernel** of a matrix $A \in \mathbb{R}^{m \times n}$:

$$\ker(A) = \left\{ x \in \mathbb{R}^n \mid Ax = 0 \right\}$$

$\hookrightarrow \vec{0}$ in $\mathbb{R}^n$

proof that it's a subspace:

let $v = \alpha v_1 + \beta v_2$ for any $v_1, v_2 \in \ker(A)$

and $\alpha, \beta \in \mathbb{R}$. Then $Av = A(\alpha v_1 + \beta v_2)$

$$\begin{array}{l} Ax = b, Av = b \\ \text{then for any } y \in \ker(A) \\ Ay = 0 \text{ so } A(v+y) = b \end{array} \quad \begin{array}{l} = \alpha Av_1 + \beta Av_2 \\ = 0 \end{array}$$

13

Column Space

Column Space

$$\mapsto A: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

The **column space** or **range** or **image** of a matrix $A \in \mathbb{R}^{m \times n}$ is another important subspace of $\mathbb{R}^m$:

$$\text{col}(A) = \{ y \in \mathbb{R}^m \mid \exists x \in \mathbb{R}^n \text{ so } Ax = y \}$$

Ex. prove that $\text{col}(A)$ is a subspace of $\mathbb{R}^m$

Definition: the **rank** of A is the dimension of $\text{col}(A)$

14

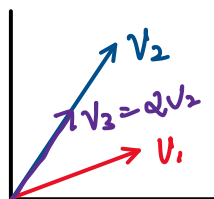
Linear Independence

A set of vectors $\{v_1, v_2, \dots, v_n\}$ is **linearly independent** if and only if:

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0$$

$$\rightarrow \alpha_i = 0 \text{ for } i = 1 \cup n$$

otherwise we call **linearly dependent**.



$\{v_2, v_3\}$ linearly dependent

$\{v_1, v_2\}$ linearly independent

Invertibility

For a square matrix $A \in \mathbb{R}^{n \times n}$, we are interested in whether $Ax = b$ has a unique solution. The following are all equivalent:

1. $\exists A^{-1} \in \mathbb{R}^{n \times n}$, such that $A^{-1}A = AA^{-1} = I = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}$

2. The columns of A are linear independent

3. The dimension of $\ker(A)$ is 0

4. $\det(A) \neq 0$

we call an invertible matrix be **non-singular**

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Eigenvalues and Eigenvectors

For a square matrix $A \in \mathbb{R}^{n \times n}$, an **eigenpair** consists of a scalar **eigenvalue** $\lambda \in \mathbb{R}$ and corresponding **eigenvector** $v \in \mathbb{R}^n$ which satisfy the following equation:

$$Av = \lambda v, v \in \mathbb{R}^n$$

Geometrically, A scales v by λ .

“Eigen” is German for characteristic

Characteristic Polynomial

The eigenvalues can be computed as solutions to the **characteristic polynomial** as follows:

$$Av = \lambda v, \rightarrow Av - \lambda v = 0 \quad \text{some } \lambda \in \text{complex!}$$
$$(A - \lambda I)v = 0$$

For non trivial solution to v , $A - \lambda I$ must be singular (not invertible). So $\det(A - \lambda I) = 0$ gives the polynomial in λ of degree n , solve it for n roots

19

Eigenspaces

The eigenvectors of $A \in \mathbb{R}^{n \times n}$ which share the same eigenvalue form a linear subspace called an **eigenspace**:

$$A(v_1 + v_2) = Av_1 + Av_2 = \lambda v_1 + \lambda v_2 = \lambda(v_1 + v_2)$$

eigenvectors

with same

eigen value λ

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Orthogonality of Vectors

Recall that two vectors $u, v \in \mathbb{R}^n$ are **orthogonal** if their dot product is zero:

$$x \cdot y = x^T y = |x| \cdot |y| \cos \theta = 0$$

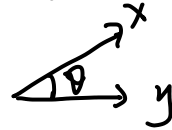
$$\cos \theta = 0 \Rightarrow \theta = 90^\circ \text{ or } 270^\circ \Rightarrow v \perp u$$

$$\therefore \cos \theta = 0$$

$$\theta = \frac{\pi}{2} \pm k\pi, k \in \mathbb{Z}$$

ie. x and y are $\perp$ to one another

$\forall$ is angle between x, y



22

Orthogonality of Matrices

Orthogonal matrices can be similarly defined:

orthogonal to one another
and of unit length

$Q \in \mathbb{R}^{m \times n}$ with orthonormal columns / rows

$$Q^T Q = \begin{bmatrix} - & q_1^T & - \\ - & q_2^T & - \\ & \vdots & \\ - & q_n^T & - \end{bmatrix} \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \dots & q_n \\ | & | & & | \end{bmatrix} = \underline{I}$$

Also: $Q^T = Q^{-1}$

23

Orthogonal matrix 每个列向量彼此长度为1且正交。且 orthogonal matrix 不改变 x 的长度，即

$$\|Ax\| = \|x\|$$